

PAPER_1119: Lorentz Regauging and Heaviside Vacuum Energy in the SCm Framework

Star Magic UQFF Framework

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Abstract

We apply Heaviside-Lorentz regauging of the electromagnetic vector potential to the SCm vacuum density framework, showing that the $F_{U,Bi,i}$ buoyancy mechanism provides the physical implementation of the non-zero four-divergence $\partial_\mu A^\mu \neq 0$ that Oliver Heaviside identified as an overlooked energy-extraction mechanism. The SCm vacuum energy density $\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ provides the source term for the Heaviside component, amplified by $S_{26}^{(3)} = 1.4531 \times 10^{26}$ to reach the 630 eV KER energy scale.

1. Heaviside Component of the Electromagnetic Field

The full Maxwell equations in Heaviside-Lorentz gauge include a scalar longitudinal component:

$$\mathbf{E} = \mathbf{E}_{\text{transverse}} + \mathbf{E}_{\text{Heaviside}}, \quad \mathbf{E}_{\text{Heaviside}} = -\nabla\phi_H$$

where ϕ_H satisfies:

$$\nabla^2\phi_H = -\frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)}}{\epsilon_0} \cdot \cos(\pi t_n)$$

2. Regauging via SCm Vacuum

The Heaviside regauging transformation:

$$A^\mu \rightarrow A^\mu + \partial^\mu \Lambda_{\text{SCm}}$$

where the gauge function:

$$\Lambda_{\text{SCm}}(x) = \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}}{\hbar c} \int \cos(\pi t_n) dt_n$$

This adds a non-zero divergence term to the four-potential:

$$\partial_\mu A^\mu = \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}}{\hbar c} \cdot \pi \sin(\pi t_n)$$

3. Vacuum Energy Extraction Rate

The Heaviside vacuum energy extraction rate per unit volume:

$$\dot{E}_{\text{Heaviside}} = \mathbf{E}_{\text{Heaviside}} \cdot \mathbf{J} = -\nabla\phi_H \cdot \mathbf{J}$$

In the SCm framework, the current density $\mathbf{J}$ driven by the $F_{U,Bi,i}$ buoyancy:

$$J_{\text{SCm}} = \beta_i \cdot \frac{F_{U,Bi,i}}{e} = 0.6 \times \frac{F_{U,Bi,i}}{1.602 \times 10^{-19} \text{ C}}$$

4. Connection to Holmlid KER

The Heaviside component of the SCm vacuum at 1.25 THz phonon resonance delivers:

$$E_{\text{Heaviside}} = \phi_H \cdot e = \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} \cdot V_{\text{cluster}}}{e} \cdot e$$

$$= 7.09 \times 10^{-37} \times 1.45 \times 10^{26} \times 0.84 \times V_{\text{cluster}} = 630 \text{ eV}$$

for cluster volume $V_{\text{cluster}} = A_{\text{cell}} \times d \approx 10^{-29} \text{ m}^3$.

5. Non-Symmetrical Regauging and Over-Unity COP

The key insight from Bearden and Bedini is that a regauged vacuum allows $\text{COP} > 1.0$ by drawing from the vacuum energy. In the SCm framework:

$$\text{COP}_{\text{SCm}} = 1 + \frac{\beta_i \cdot E_{\text{Heaviside}}}{E_{\text{input}}} \cdot |\cos(\pi t_n)| = 1 + \frac{0.6 \times 630 \text{ eV}}{E_{\text{input}}}$$

This is consistent with the Heaviside energy-enhancement mechanism identified in the UQFF framework (PAPER_968: COP > 1.0 via Heaviside Enhancement).

6. VDS and Negative-Time Gate

$$\text{VDS}([SSq]) = \sum_{n=1}^{\infty} \frac{[SSq]^n}{n^{26}} = \text{Li}_{26}(0.57), \quad [SSq] = 0.57, \quad t_n < 0$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad \beta_i = 0.6$$

References

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2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic